

The cumulative cost of additional wakefulness: dose-response effects on neurobehavioral functions and sleep physiology from chronic sleep restricti...

OBJECTIVES: To inform the debate over whether human sleep can be chronically reduced without consequences, we conducted a dose-response chronic sleep restriction experiment in which waking neurobehavioral and sleep physiological functions were monitored and compared to those for total sleep deprivation. **DESIGN:** The chronic sleep restriction experiment involved randomization to one of three sleep doses (4 h, 6 h, or 8 h time in bed per night), which were maintained for 14 consecutive days. The total sleep deprivation experiment involved 3 nights without sleep (0 h time in bed). Each study also involved 3 baseline (pre-deprivation) days and 3 recovery days. **SETTING:** Both experiments were conducted under standardized laboratory conditions with continuous behavioral, physiological and medical monitoring. **PARTICIPANTS:** A total of $n = 48$ healthy adults (ages 21-38) participated in the experiments. **INTERVENTIONS:** Nocturnal sleep periods were restricted to 8 h, 6 h or 4 h per day for 14 days, or to 0 h for 3 days. All other sleep was prohibited. **RESULTS:** Chronic restriction of sleep periods to 4 h or 6 h per night over 14 consecutive days resulted in significant cumulative, dose-dependent deficits in cognitive performance on all tasks. Subjective sleepiness ratings showed an acute response to sleep restriction but only small further increases on subsequent days, and did not significantly differentiate the 6 h and 4 h conditions. Polysomnographic variables and delta power in the non-REM sleep EEG—a putative marker of sleep homeostasis—displayed an acute response to sleep restriction with negligible further changes across the 14 restricted nights. Comparison of chronic sleep restriction to total sleep deprivation showed that the latter resulted in disproportionately large waking neurobehavioral and sleep delta power responses relative to how much sleep was lost. A statistical model revealed that, regardless of the mode of sleep deprivation, lapses in behavioral alertness were near-linearly related to the cumulative duration of wakefulness in excess of 15.84 h (s.e. 0.73 h). **CONCLUSIONS:** Since chronic restriction of sleep to 6 h or less per night produced cognitive performance deficits equivalent to up to 2 nights of total

sleep deprivation, it appears that even relatively moderate sleep restriction can seriously impair waking neurobehavioral functions in healthy adults. Sleepiness ratings suggest that subjects were largely unaware of these increasing cognitive deficits, which may explain why the impact of chronic sleep restriction on waking cognitive functions is often assumed to be benign. Physiological sleep responses to chronic restriction did not mirror waking neurobehavioral responses, but cumulative wakefulness in excess of a 15.84 h predicted performance lapses across all four experimental conditions. This suggests that sleep debt is perhaps best understood as resulting in additional wakefulness that has a neurobiological "cost" which accumulates over time.